

11. Expansion-Modulus

Contours: Thermal Stress/ $^{\circ}\text{K}$ (MPa/K)

MFA:88-91

$\alpha E = 0.1$
(MPa/K)

1.0

10

Linear Expansion Coefficient α (10^{-6}K^{-1})

Polymer
Foams

Engineering
Polymers

Elastomers

Cork

LDPE

HDPE

PP

Nylons

PS

Epoxies

PC

PMMA

Polyesters

MEL

Ice

Balsa

$\perp$ to
Grain

Oak

Pine

Ash

Wood
Products

Woods

Composites

Engineering
Alloys

Pb

Mg

Zn

Al

Alloys

Cu

Steels

Cast
Irons

Ni

Be

Al₂O₃

ZrO₂

W

SiC

Sialons

GFRP

Laminates

Cement
Concrete

Pottery

Uniply

CFRP

Common
Rocks

S-Glass

B-Glass

Porous
Ceramics

SiO₂

Engineering
Ceramics

Invar

αE (MPa/K) = 0.01

0.1

Youngs Modulus, E (GPa)

0.01 0.1 1.0 10 100 1000

